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NIELIT

Intelligent Urban Traffic light Management System Using Reinforcement Learning and Deep Learning

M.Tech 2^{nd} Semester Mini project

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INTRODUCTION

The Core Issue: Rapid urbanization in cities like Bangalore leads to severe traffic congestion, increasing travel time, fuel consumption, and greenhouse gas emissions.

Limitations of Current Systems: Traditional traffic lights operate on fixed timers. They cannot adapt to real-time, dynamic traffic fluctuations (e.g., peak hours, emergency situations).

PROBLEM STATEMENT

The Urban Mobility Crisis

Fixed-Time Actuation

Traditional formula fails in dynamic, heterogeneous Indian traffic.

Emergency Fatalities

Lack of intelligent preemption causes severe ambulance delays.

Idling Emissions

Sub-optimal stop-and-go waves increase and CO2 exponentially.

OBJECTIVES

1. **AI Control:** To develop a self-learning Reinforcement Learning (PPO) agent that dynamically adjusts traffic signal phases in real-time.
2. **Emergency Preemption:** To drastically reduce ambulance waiting times by creating intelligent green-wave priorities.
3. **Eco-Routing (Pollution Control):** To minimize CO2 and PM2.5 emissions by reducing stop-and-go traffic waves using HBEFA3 emission models.
4. **Real-World Validation:** To validate the AI model using actual macroscopic traffic data (Kaggle) mapped to a microscopic simulation environment (SUMO).

Literature Review (Part 1)

Reference / Focus Area	Year	Problem Identified	Model used	Key Limitations
Optimised Traffic Light Management Through reinforcement Learning	2022	Fixed timers cause delays delays	Q-Learning (Basic (Basic RL)	Fails in complex, multi-lane intersections due to state-space
Optimizing Traffic Flow With Reinforcement Learning: A Study on Traffic Light Management	2024	Dynamic traffic needs adaptive signals	Deep Reinforcement learning	Better than fixed timers, but lacked real-world world map integration.
DQN- Based Traffic Signal Control Systems	2021	Traditional Tabular Q – learning cannot handle the massive, continuous state space of real-world urban traffic grids.	Deep Q-Networks (DQN)	While effective for single intersections, basic basic DQN models often struggle with convergence and can overestimate values, values, leading to sub-optimal traffic light light phases.
Intelligent / Dynamic Traffic Signal Control using Advanced Machine Learning	2025	Accurately measuring traffic traffic volume in real time is time is difficult with traditional sensors.	Computer Vision(YOLO/ CNN) for object Detection	Focuses primarily on the “vision” aspect(counting cars) but relies on basic rule- rule- based logic to switch lights, missing a missing a true RL decision-making agent. agent.
Deep Learning Flow Prediction Prediction	2021	Non – linear traffic patterns patterns are hard to capture.	LSTM / GRU	Excellent at predicting *when* congestion will occur, but lacks a control mechanism to stop it.

Literature Review (Part 2)

Reference / Focus Area	Year	Problem Identified	Model / Approach Used	Key Limitations
Dynamic traffic light control system based on real- time traffic density	2023	Pre-timed traffic signals cause cause unnecessary waiting when lenses are empty.	IOT & Hardware Sensors(IR/ Ultrasonic)	Highly hardware- dependent. Uses hardcoded "If-Else" rules rather than a self learning, adaptive AI.
Cooperative Traffic Signal Control of n- intersections Using a Double Deep Q- Network Agent	2024	Traffic controllers act in isolation. Solving one intersection often pushes the traffic jam to the next adjacent intersection.	Double Deep Q-Network(DDQN)	The Complex reward sharing sharing mechanism between between multiple intersections makes the model model computationally heavy heavy and difficult to train efficiency.
Multi-agent Traffic Light Controls with SUMO	2023	Comparing the effectiveness of deep learning agents versus simple rule-based agents in highly unpredictable environments.	DQN Agents vs Reactive Agent in SUMO	The DQN actually struggled to learn an optimal policy due to the highly non-stationary nature of traffic, proving that RL models require highly refined state-reward definitions to work well.

DATA ACQUISITION AND FEATURE ENGINEERING

1. The Raw Dataset: Bangalore City Traffic Pulse
Source: Open-source Kaggle Dataset capturing macroscopic traffic patterns across major Bengaluru IT hubs.
Characteristics: Contains holistic data including volume, speeds, congestion levels, and emissions for various intersections.
https://www.kaggle.com/datasets/preethamgouda/banglore-city-traffic-dataset

2. Selective Feature Extraction (Simulation Inputs) To build a realistic simulation environment without biasing the AI, only **three primary macroscopic features** were extracted:
Area Name: Used to geographically isolate and filter our specific case study zones (e.g., Indiranagar, Koramangala).
Road/Intersection Name: KFC Junction and Sony World Junction.
Traffic Volume: The core input metric representing the raw vehicle count passing the location per day.

METHODOLOGY: MACRO-TO-MICRO PIPELINE

1. Macro Data Extraction

Processed raw Bengaluru Traffic Pulse dataset from Kaggle. Extracted temporal volume distribution for target junctions.

2. Micro-Scaling

Converted aggregated hourly flow (e.g., 61,000 veh/hr) into probabilistic microscopic spawn rates (~1.17s per vehicle insertion) to prevent gridlock logic errors.

3. SUMO

Integrated XML route networks with **SUMO (Simulation of Urban MObility)**, configuring heterogeneous agents (Cars, Autos, Buses, Ambulances) via TraCI.

How It Works – The RL Architecture

State (Input)

Current traffic conditions extracted from SUMO (e.g., number of waiting vehicles, queue length in each lane).

Action(Output)

The agent decides the next optimal traffic light phase(e.g.. Switch to Green for North-South).

Reward

The agent receives a positive reward for reducing cumulative wait times, and a penalty for increasing congestion.

PPO ALGORITHM

Proximal Policy Optimization (Actor-Critic)

Selected PPO over DQN due to its **Trust Region constraint**. It prevents the AI from making drastic, unsafe changes to traffic lights that would collapse the network.

Hyperparameter Tuning Engine:
Learning Rate: 0.0003
Clip Range (ε): 0.2 (Ensures stability)
Entropy Coef: 0.05 (Forces state-space exploration)

Q-Learning / DQN Limitations:

Discrete action spaces struggle with continuous multi-intersection traffic states.

- **Catastrophic Forgetting:** DQN models become unstable under extreme congestion or weather shifts.
- Lack of Multi-Objective optimization (combining eco-routing with emergency priority).

MULTI-OBJECTIVE REWARD ENGINEERING

To prevent **Reward Hacking** (where the agent keeps one lane green indefinitely), a Time-Accumulated Penalty function was engineered combining Traffic Flow, Emissions, and Preemption.

$$R_t = - \sum (W_{normal}) - \sum (W_{ambulance} \times 100) - (CO2_{HBEFA3} \times 0.5)$$

Time-Accumulated Delay

Penalizes the agent exponentially as vehicle wait-time increases, forcing cycle switching.

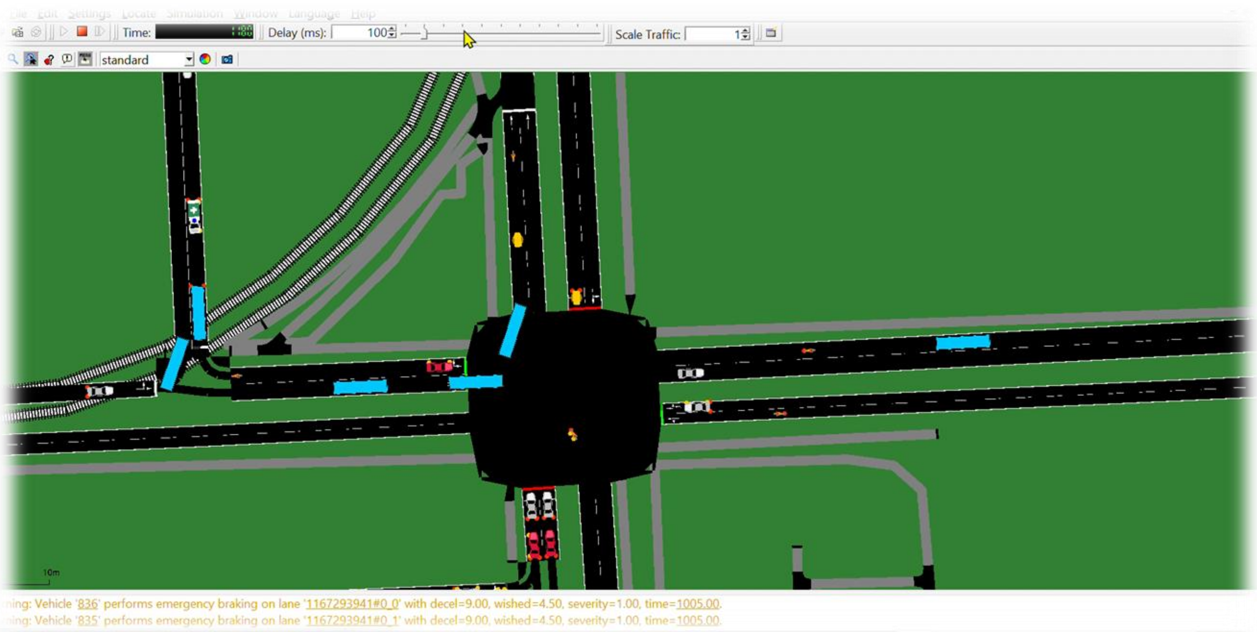
Emergency Priority

A 100x multiplier penalty for halting ambulances, guaranteeing immediate green-wave actuation.

Eco-Routing

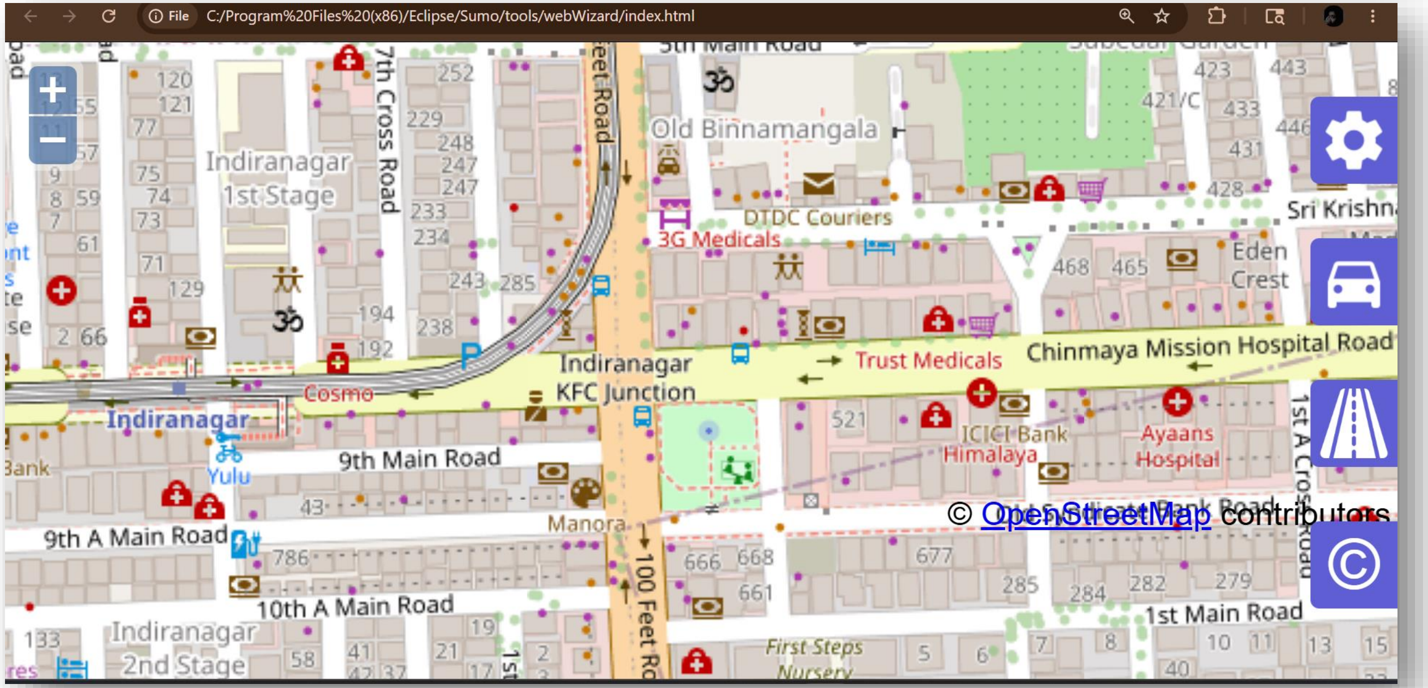
Incorporated real-time HBEFA3 emission models to minimize high-acceleration stop-and-go waves.

CASE STUDY 1:BANGALORE INDIRANAGAR (KFC JUNCTION)



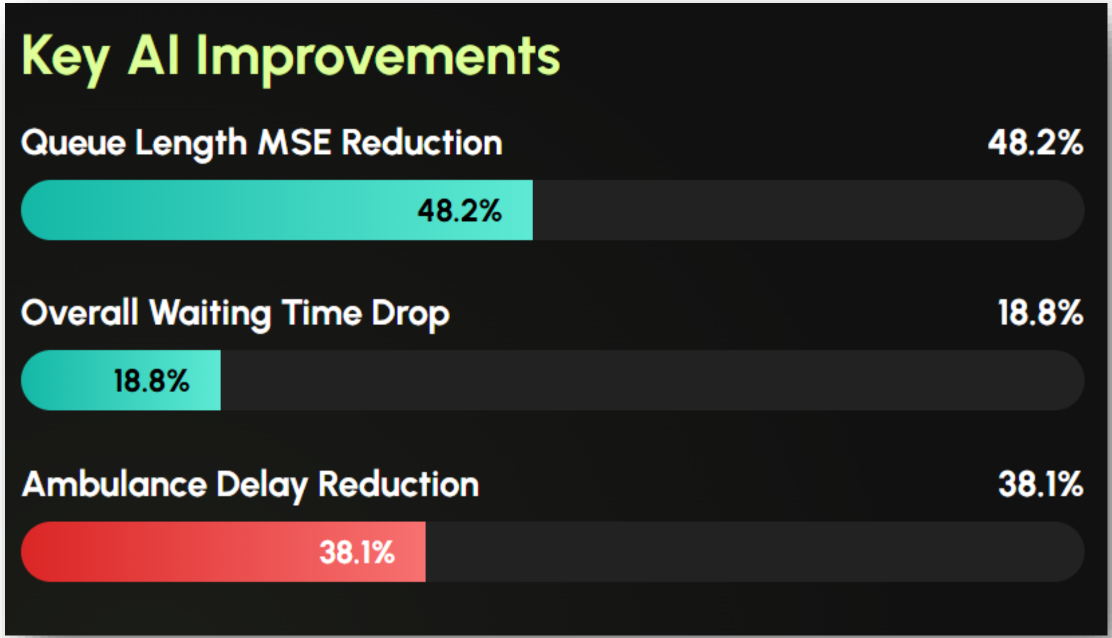
Network Characteristics

Standard urban intersection with moderate cross-traffic density and intermittent emergency vehicle presence.



CASE STUDY 1: RESULT

Validation
The PPO model successfully stabilized the intersection, shifting the system from heavy queue formation (Baseline MSE: 1049.9) to a continuous flow state (RL MSE: 543.5).

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🎓 M.TECH THESIS RESULTS CALCULATOR 🎓
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📊 1. OVERALL WAITING TIME
   - Without RL (Baseline): 757.02 seconds
   - With RL Model:         614.39 seconds
   ✅ Improvement:          18.84% reduction in traffic delay

🚑 2. EMERGENCY VEHICLE (AMBULANCE) DELAY
   - Without RL (Baseline): 35.67 seconds
   - With RL Model:         22.05 seconds
   ✅ Improvement:          38.18% faster emergency response

🍃 3. CO2 EMISSIONS (Eco-Routing)
   - Without RL (Baseline): 340.70 grams
   - With RL Model:         283.20 grams
   ✅ Improvement:          16.88% reduction in pollution

📉 4. QUEUE LENGTH (MSE from Ideal State = 0)
   - Baseline MSE: 1049.99
   - RL Model MSE: 543.54
   (Lower MSE means the queue is closer to zero, which is better!)

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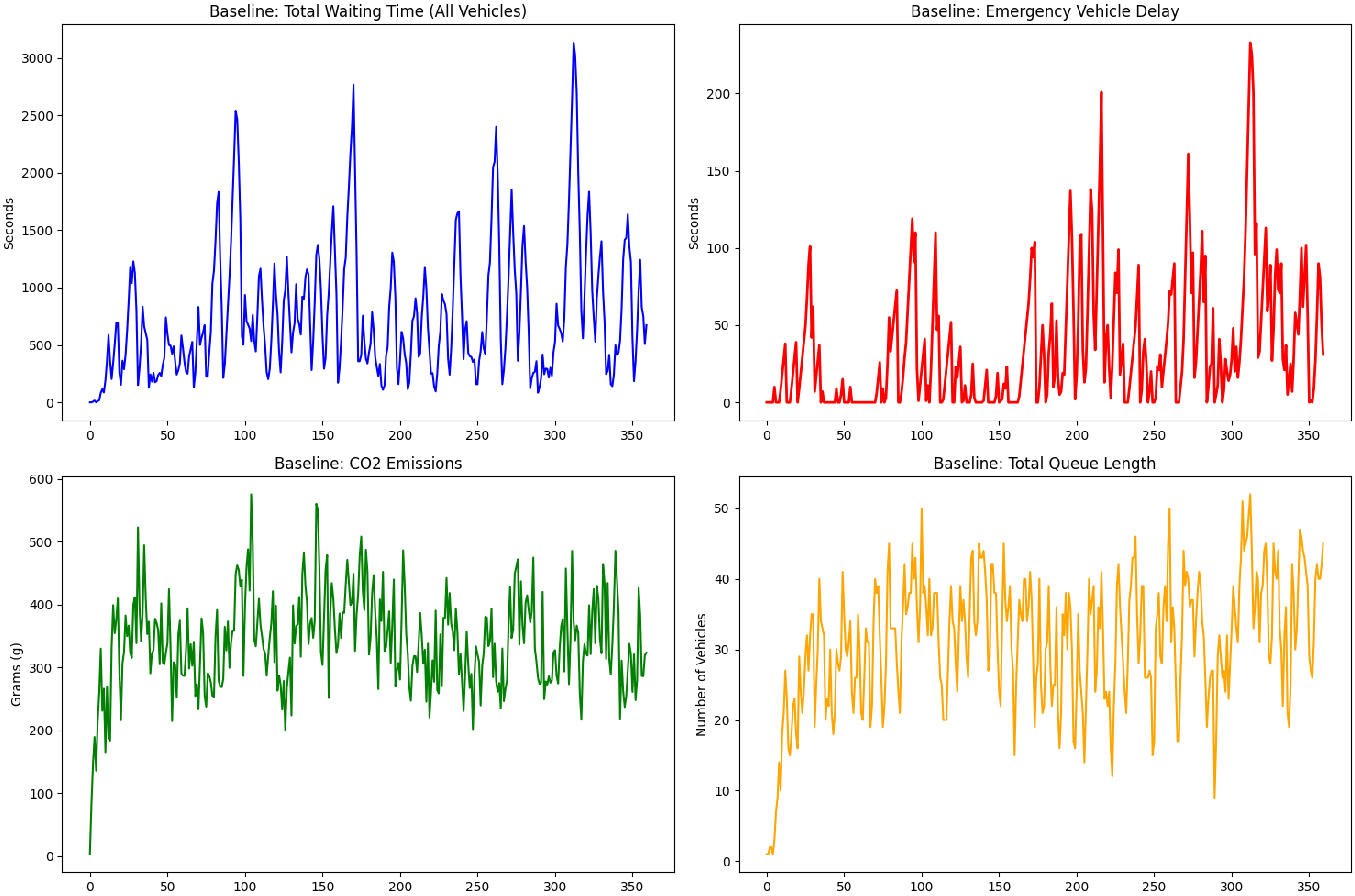
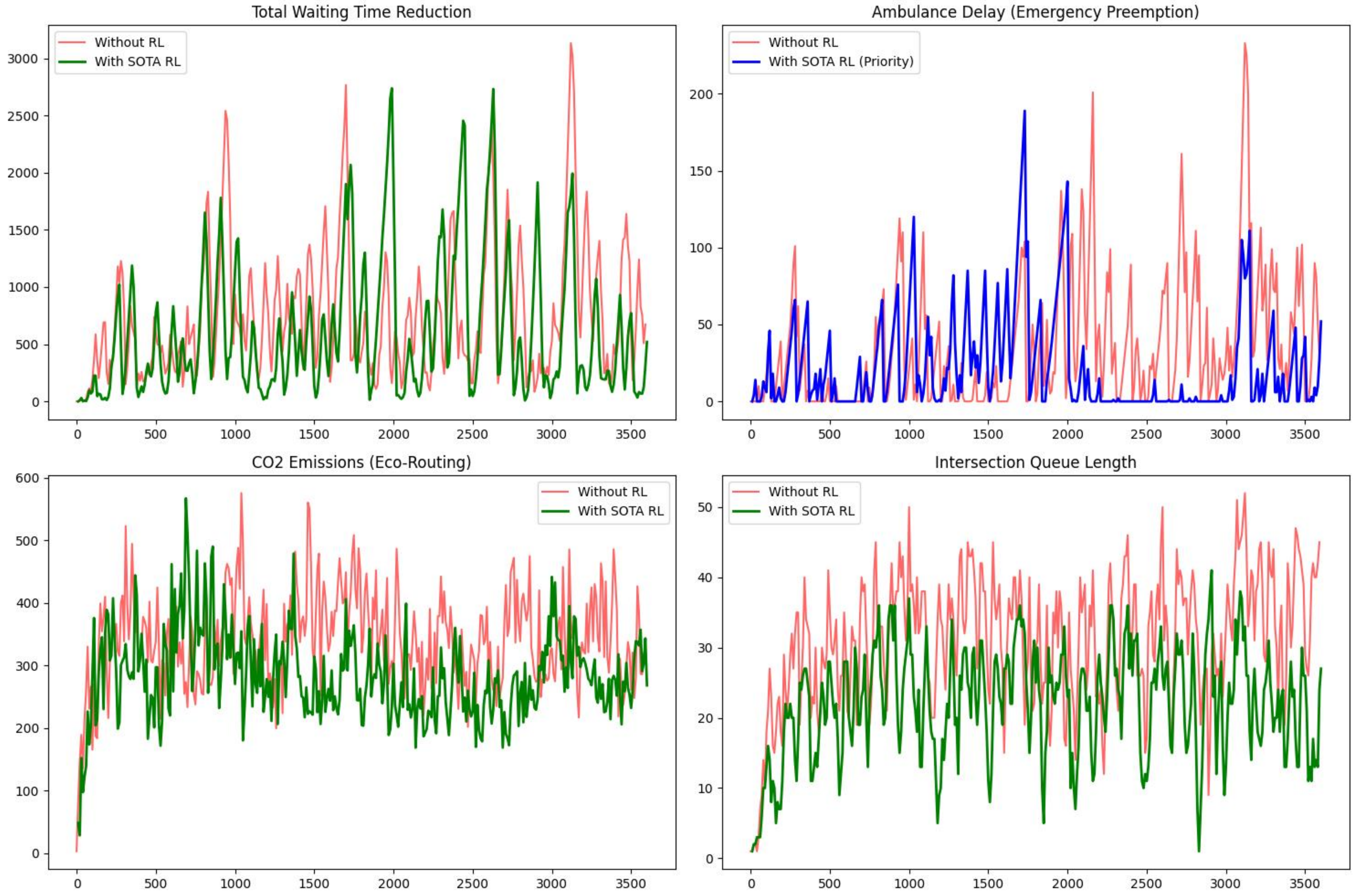
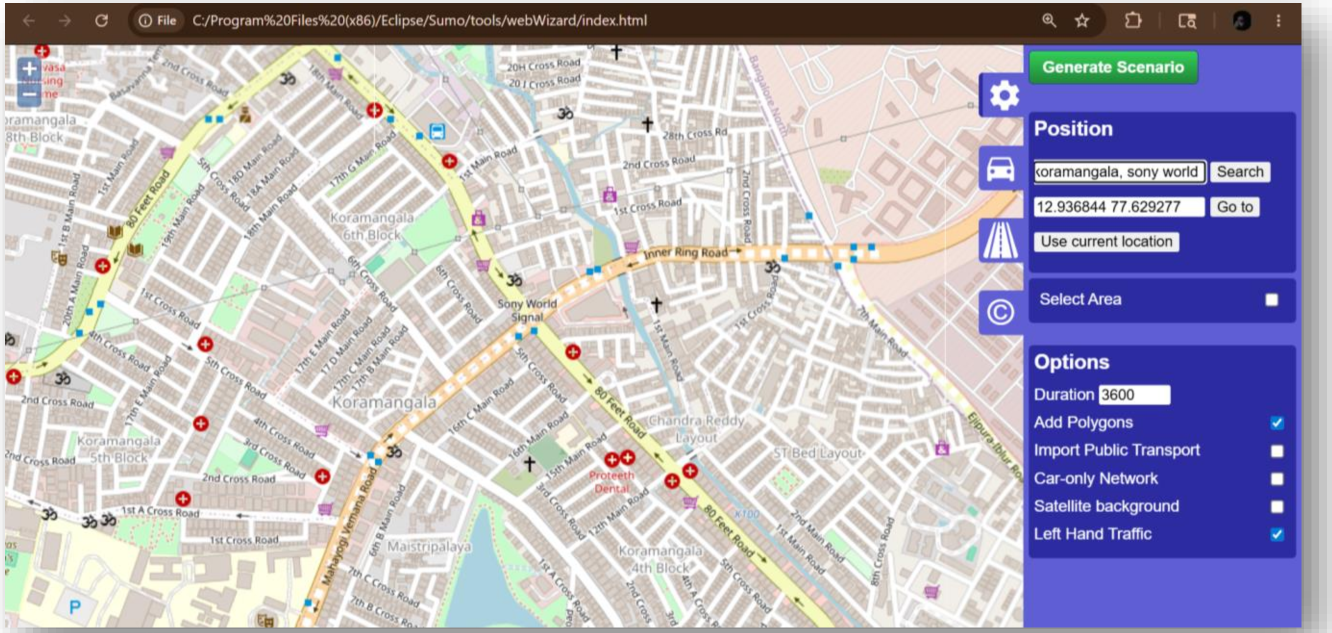


Baseline: Total Waiting Time (All Vehicles)
Seconds
Baseline: Emergency Vehicle Delay
Seconds
Baseline: CO2 Emissions
Grams (g)
Baseline: Total Queue Length
Number of Vehicles



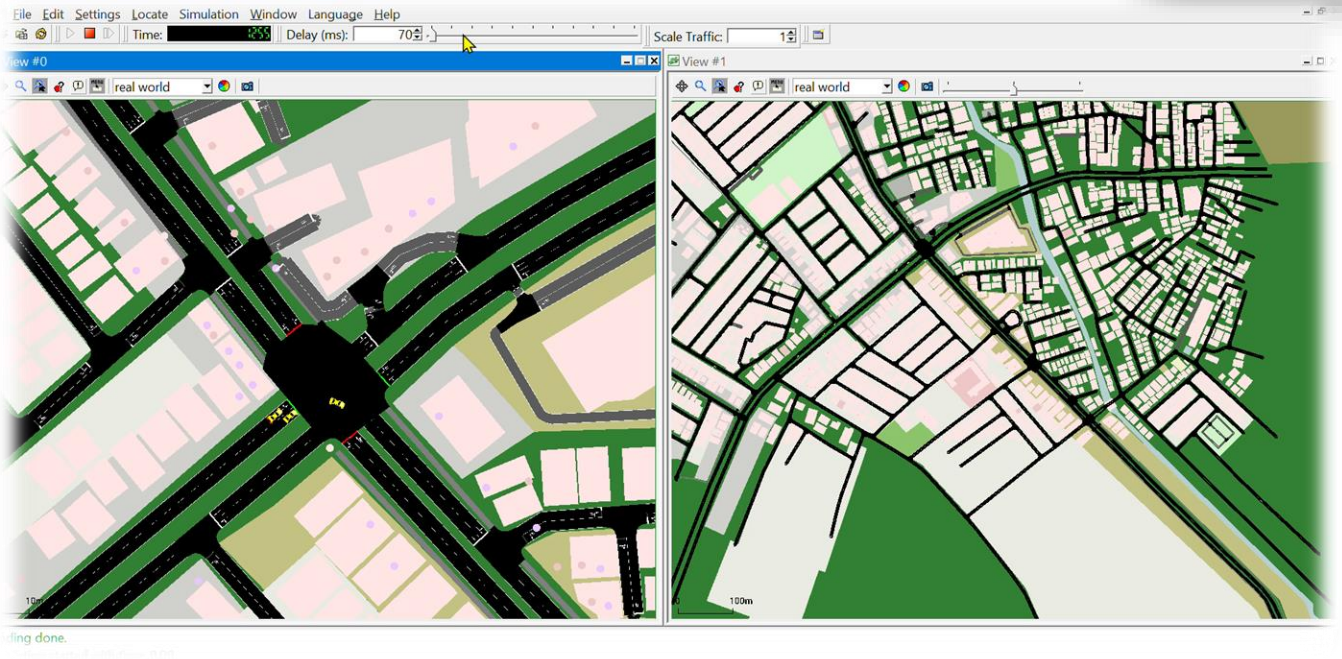
Total Waiting Time Reduction
Without RL
With SOTA RL
0
500
1000
1500
2000
2500
3000
3500
Ambulance Delay (Emergency Preemption)
Without RL
With SOTA RL (Priority)
0
50
100
150
200
CO2 Emissions (Eco-Routing)
Without RL
With SOTA RL
100
200
300
400
500
600
Intersection Queue Length
Without RL
With SOTA RL
10
20
30
40
50

CASE STUDY 2:BANGALORE KORAMANGALA (SONY WORLD JUNCTION)



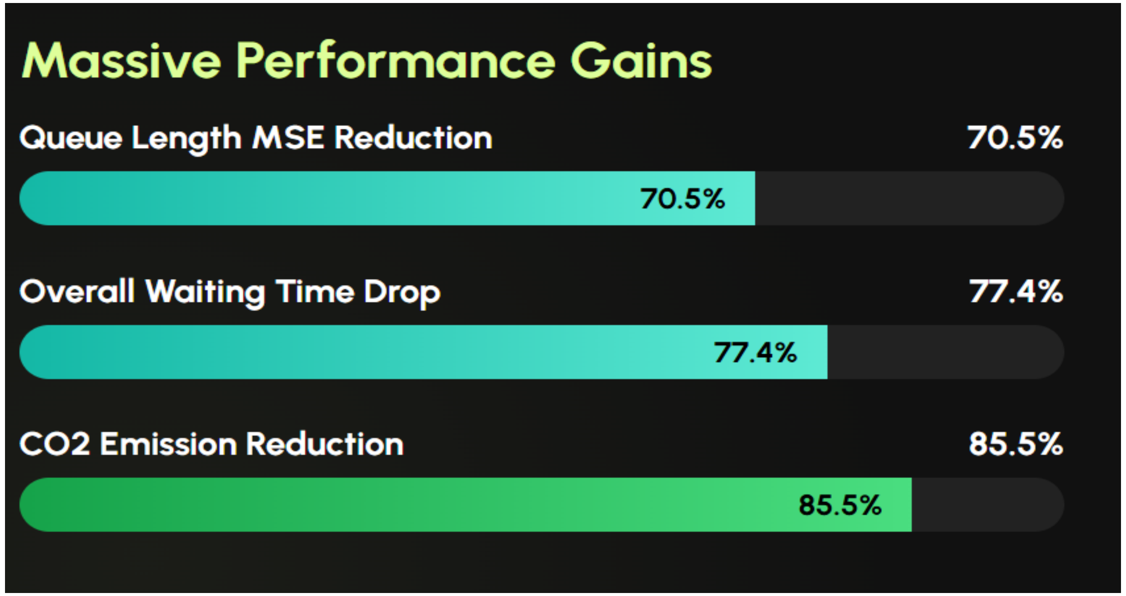
Extreme Congestion Stress Test

To test generalization, the model was deployed on Sony World Junction—a highly asymmetric, severely bottlenecked network.



CASE STUDY 2: RESULT

Generalization Proof
Wait Time: Dropped from 789s to 177s.
Ambulance Clearance: 57.3% faster clearance.
Emissions: Reduced from 2760 kg to 398 kg.
Insight: The worse the original fixed-timer setup, the greater the PPO agent's relative impact.

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   🚦 AI TRAFFIC OPTIMIZATION RESULTS (SONY WORLD)
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1. QUEUE LENGTH (Traffic Jam)
   - Without AI : 2232.03 vehicles (Average)
   - With PPO AI: 1153.11 vehicles (Average)
   => IMPROVEMENT : 48.34% Reduce hua!

2. QUEUE MSE (Mean Squared Error - Penalizing large jams)
   - Without AI : 6216916.55
   - With PPO AI: 1833242.37
   => MSE REDUCTION : 70.51% Reduce hua!

3. AVERAGE WAITING TIME (Sabhi gaadiyon ka)
   - Without AI : 789.45 Seconds
   - With PPO AI: 177.85 Seconds
   => IMPROVEMENT : 77.47% Time bacha!

4. AMBULANCE WAITING TIME (Emergency Priority)
   - Without AI : 972.69 Seconds
   - With PPO AI: 415.04 Seconds
   => IMPROVEMENT : 57.33% Faster Clearance!

5. TOTAL CO2 EMISSIONS (Pollution)
   - Without AI : 2760.84 kg
   - With PPO AI: 398.02 kg
   => IMPROVEMENT : 85.58% Pollution kam hua!
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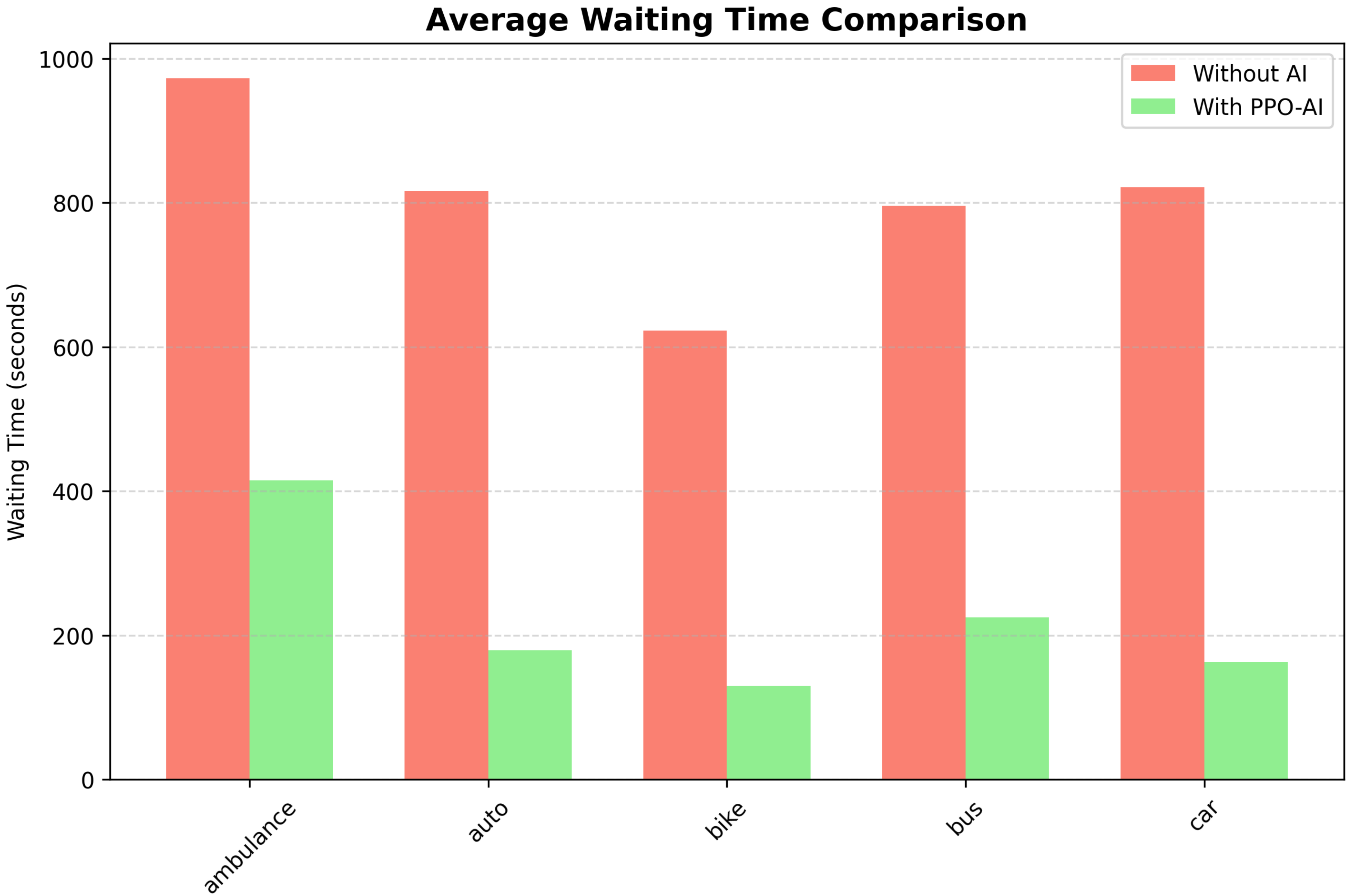
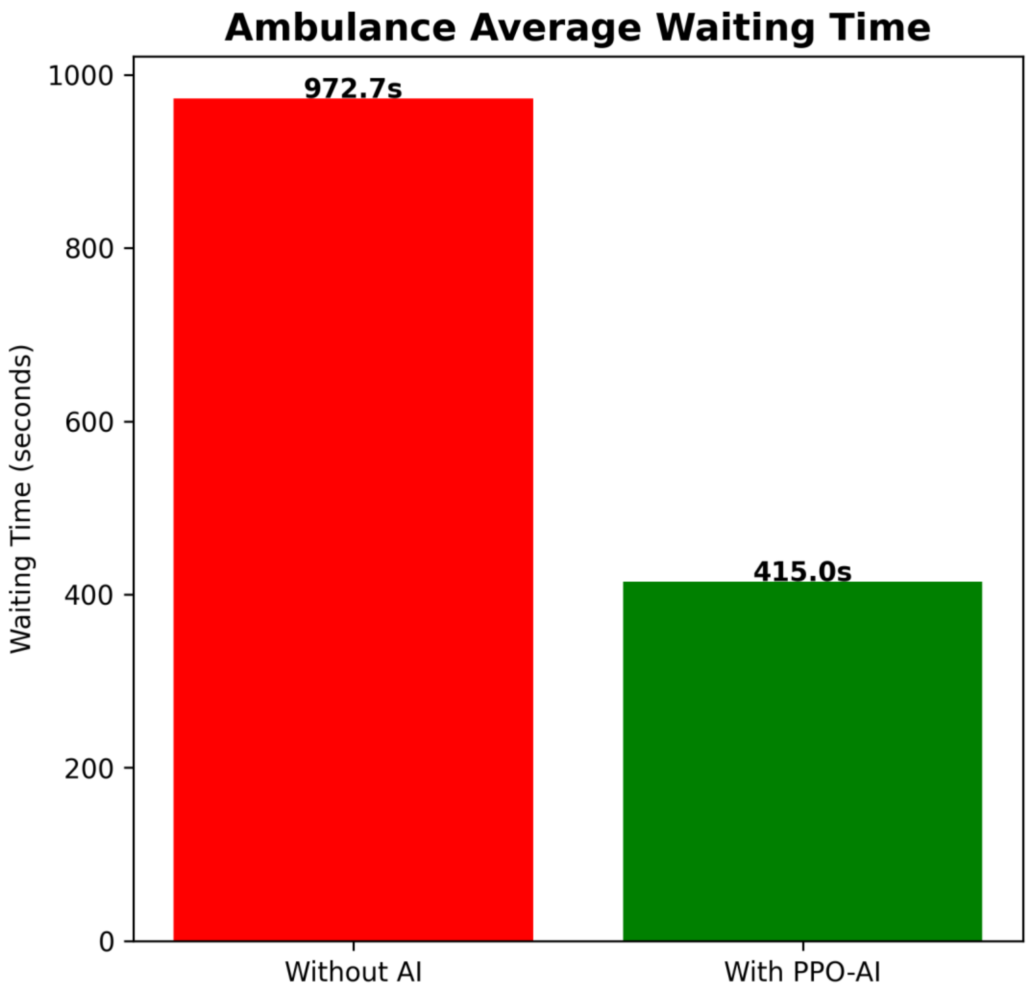
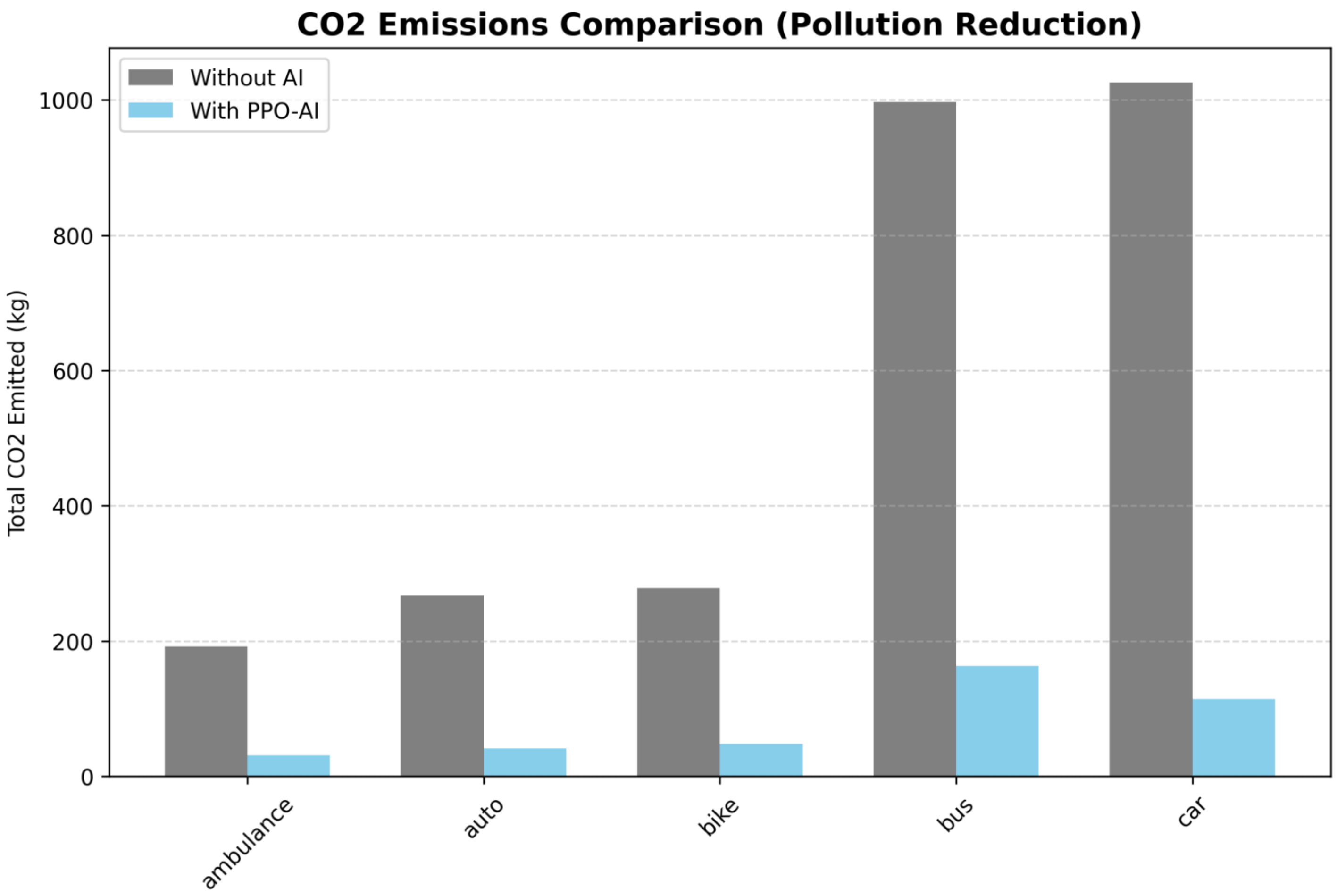


Average Waiting Time Comparison
Without AI
With PPO-AI
Waiting Time (seconds)
1000
800
600
400
200
0
ambulance
auto
bike
bus
car


Ambulance Average Waiting Time
972.7s
415.0s
Waiting Time (seconds)
0
200
400
600
800
1000
Without AI
With PPO-AI


CO2 Emissions Comparison (Pollution Reduction)
Without AI
With PPO-AI
Total CO2 Emitted (kg)
0
200
400
600
800
1000
ambulance
auto
bike
bus
car

CONCLUSION

The integration of Deep Reinforcement Learning (PPO) with real-world macro data proves highly superior to traditional fixed-cycle controllers. The agent successfully learned **Indirect Optimization**—minimizing massive queues by targeting only Wait-Time and Emissions.

Research Achievements

- Resolved Reward Hacking loops.
- Achieved 57% faster Emergency response.
- Validated across multiple network topologies.

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Thank you